SIEMENS

Model FN2017-U1, FN2018-U1

Ethernet Module (Multi Mode Fiber Optic / Single Mode Fiber Optic)

Installation

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2026-02-06

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1 Introduction

The optional models FN2017-U1 Ethernet Module (MM) and FN2018-U1 (SM) are media adapters which provide interconnecting fiber optic links for networked systems.

The Operating Temperature Range and Humidity is 0 to 49 °C, and 93% at 32 °C. Install in accordance with the National Electrical Code, ANSI/ NFPA 70, the Standard for Installation and Classification of Burglar and Holdup Alarm Systems, UL 681, and the Standard for Central-Station Alarm Services, UL 827, as applicable.

2 Features

The principal features of the FN2017-U1 (MM) and FN2018-U1(SM) include:

- Easily plugs into the host card
- Automatically identified by the system at power up
- Can be configured as Class B (DCLB) or Class X (DCLC)

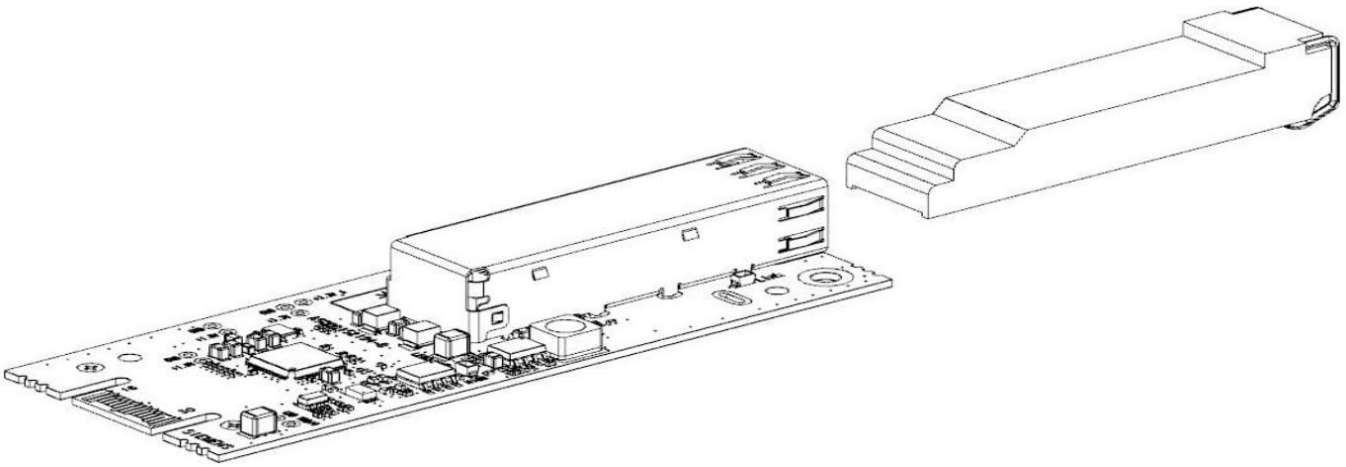


Fig. 1: FN2017-U1/FN2018-U1 Ethernet Module with removable fiber media

3 Pre-Installation

The fiber optic media adapters, either single-mode or multi-mode, come pre-installed into the FN2017-U1 and FN2018-U1. There are no configuration switches or jumpers on the FN2017-U1 or FN2018-U1.

4 Operation

The FN2017-U1and FN2018-U1 can be used in both the Desigo CC Modular and the Cerberus Pro Modular systems and are compatible with the XINC Network Ring Card. The XINC provides two sockets for media adapters such as the FN2017-U1and FN2018-U1 and they can be installed into either socket. Mixing of media adapters, such as the FN2016-U1, FN2017-U1, and FN2018-U1 media adapters on the XINC is allowed depending on the requirements of a specific installation. While the mixing of media adapter types on a given XINC is allowed they must be matched at each end of the connection between XINCs. Refer to XINC Installation Instructions, Document ID A6V12412368, for details regarding network configuration, media adapter installation and wiring instructions.

5 Wiring

The fiber optic cables use Standard LC connectors.

6 Electrical Rating

FN2017-U1	49mA (max) @ 24VDC (100Mbps)
FN2018-U1	40mA (max) @ 24VDC (100Mbps)

Table 1: Electrical Ratings

7 Output Signal Attenuation

As a signal travels down an optical fiber, its level is reduced due to absorption by the fiber. This sets the maximum length of an optical connection. The actual length is determined by the transmitter output level, the length of the fiber, and the receiver sensitivity. The maximum length is reached when the fiber attenuation is the same as the difference between the transmitter output and the receiver sensitivity less than 3 dBm for an error margin. These values are different depending upon the module type used and the kind of optical fiber which is used to connect the modules.

The following table gives the maximum allowable signal attenuation for the different kinds of fiber used with each type of module. It also gives the maximum length of an optical link using that combination of module and fiber and are indicative only. Signal attenuation due to fiber loss, termination quality and splices determines maximum link length.

Module Type	Fiber Type	Signal Attentuation	Link Length
FN2017-U1 (MM)	50/125 µm 62.5/125 µm	7.5 dB 11 dB	2.0 km 2.0 km
FN2018-U1 (SM)	9/125 µm	30 dB	40 km

Table 2: Ethernet Module Output Signal Maximum Attenuation and Link Length

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